HIREFLOW AI — QUALITY ENGINE & ARCHITECTURE MANUAL

The Comprehensive Technical Reference to All Subsections, UI/UX Layouts, Project Briefs & Calibration Engines

VERSION 4.0.0 | 106 SUITES PASSED (738 TESTS) | PRODUCTION ARCHITECTURE EDITION

Chapter 1: System Architecture & UI/UX Design System

HireFlow v4 is engineered as an offline-capable, zero-egress candidate career decision and evidence platform. It operates 100% inside the browser with zero external telemetry, zero server-side state, and deterministic pure engines.

1.1 Dark Ambient Canvas & Design Tokens

- **Background Canvas**: Deep slate/neutral tones (`#020617` and `#0a0a0a`) paired with subtle ambient radial gradients.
- **Glassmorphic Surfaces**: `bg-surface/70` with `backdrop-blur-md` and `border-border/80` for tactile elevation.
- **Color Accent Tokens**:
  - **Sky / Blue (#0284c7)**: Primary action controls, active resume chips, and interactive navigation.
  - **Emerald / Green (#059669)**: High-quality evidence, verified credentials, and L3 measured outcomes.
  - **Amber / Yellow (#d97706)**: Medium priority items, run-on sentences, and needs-recheck badges.
  - **Red / Rose (#dc2626)**: Weak action verbs, unverified claims, formatting dangers, and skill gaps.
- **Typography**: High-density system sans-serif hierarchy; monospace (`font-mono`) for metrics, code tokens, and math formulas.

1.2 Component Hierarchy & Persistent Navigation

Every view in HireFlow follows a disciplined component tree:

- **PublicNavbar / PageNav**: Sticky top navigation displaying logo, quick tool links, the global `ProfileResumeChip` ("Using: <resume> (change)"), and `OfflineIndicator`.
- **Tool Workspace**: Structured into Hero Control Bar → Responsive Multi-Column Grid → Live Output Canvas.
- **Slide-Over & Bottom-Sheet Drawers (`Drawer.tsx`)**: `w-96` to `w-[480px]` slide-over on desktop; automatically converts to an `h-[85vh]` swipeable bottom sheet on mobile viewports (<768px).
- **Global Floating Navigator (`LocalhostNavigator.tsx`)**: Triggered via `Ctrl + /` for rapid keyboard switching across all 14 tools and views.

Chapter 2: Comprehensive Subsection-by-Subsection UI/UX Breakdown

2.1 ATS Resume Roaster & Character-Span Annotator (`/tools/resume-roaster`)

> **Core Subsections & Interaction Canvas**
>
> - **Score Header**: Large circular ATS Score gauge (0-100), 4 calibrated pillar badges (Hard Skills, Evidence, Fresher, Format), and target alignment badge. Includes `(i)` ExplainDrawer trigger.
> - **3-Way View Mode Switcher**:
>   - `Annotated View`: Text highlighted with character-span spans (**Red weak verbs** , **Amber passive/run-on** , **Green metrics**).
>   - `Clean View`: High-contrast typography view with inline score badges.
>   - `Raw View`: Monospace text stream showing exactly what the parser parsed (reveals column mangling).
> - **Span Drawer & Rewrite Applicator**: Clicking any highlighted span slides out a drawer detailing screener rationale and offers a 1-click rewrite replacement.
> - **TabBar Subsections**: Overview, Skills Breakdown, Fix These First (prioritized by score impact), Learning Path, and Annotated View.

2.2 Dual-Role Job Description Compare Engine (`/tools/jd-compare`)

> **Core Subsections & Interaction Canvas**
>
> - **Dual Input Zones**: Side-by-side textareas for Role A and Role B with quick-fill from 7 Indian Fresher Role Presets.
> - **Overlap Breakdown**: Displays shared requirements, unique Role A skills, and unique Role B skills.
> - **Radar Comparison**: Visual radar chart comparing candidate coverage against both roles simultaneously.
> - **Deterministic Gap Derivation**: Automatically extracts missing skills into the global profile gap list.

2.3 Handmade Resume Tailor & Studio Canvas (`/tools/tailor`)

> **Core Subsections & Interaction Canvas**
>
> - **Left Column (Handmade Resume Editor)**: 2-column cards for Profile, Summary, Experience, Projects, Skills, and Education. Includes live requirement mapping, in-place bullet elevation ("Elevate Bullet" button), and drag-and-drop reordering.
> - **Right Column (Live Paper Canvas - `DarkPaperPreview.tsx`)**: Real-time A4 preview with live sync, responsive zoom, and the 1-Page Overflow Sentinel (warns if text spills onto page 2; includes `Auto-Fit One Page` adjuster).
> - **Pre-Export Formatting Sentinel (D15)**: Scans for broken Unicode glyphs, widow/orphan lines (< 3 words), and heading overflows before downloading PDF.

2.4 Project Build Briefs & Set-Cover Bundler (`/tools/build-briefs`)

> **Core Subsections & Interaction Canvas**
>
> - **Domain Category Pills**: 14 color-coded domains (Distributed Systems, Web, Data Eng, AI/ML, GenAI, DevOps, Security, FinTech, etc.).
> - **Greedy Set-Cover Bundler Tray**: Calculates the minimum combination of projects covering all open candidate gaps.
> - **Brief Cards & Detail Modal**: Detailed view of all 10 deep sections (Problem Statement, Use Cases, Architecture, Scale Targets, Screener Rubric, Resume Bullet Formula).
> - **JSZip Starter Scaffold Generator (D17)**: Generates starter repository ZIP in-browser (README with Mermaid diagram, config, tests).
> - **Resume Ledger Injection**: Injects completed project bullet with `[METRIC]` placeholders directly into draft.

2.5 Credential & Opportunity Engine (`/tools/credential-engine`)

> **Core Subsections & Interaction Canvas**
>
> - **Category Subsections**: Certifications, Events (Hackathons, CTFs, Contests), Open Source (GSoC, LFX), Opportunity Plan.
> - **Opportunity Detail Modal**: Exam dates, syllabus, official links, fee waiver eligibility, and provenance badges.
> - **RFC 5545 iCalendar Export (D18)**: Generates `.ics` calendar file with exam dates and milestone study schedules.
> - **National Exam Checker**: Rule-based eligibility checker for GATE, ISRO ICRB, AWS SAA, CDAC.

2.6 Career Trajectory Hub & What-If Simulator (`/tools/career-path`)

> **Core Subsections & Interaction Canvas**
>
> - **Career Explorer**: Leveling tracks across IT Services, Unicorns, GCCs, and Big Tech India with compensation visuals.
> - **What-If Simulator (D20)**: Interactive sliders for CGPA (5.0-10.0), College Tier (1/2/3), Branch, and Internships. Displays campus corporate cutoff gates (e.g. 7.0 CGPA) and reveals exact gap prerequisites without fake odds.

Chapter 3: Project Build Briefs Architecture, Schema & Quality Rubric

3.1 The 15-Field Project Brief Schema

Every project brief in `src/data/suite/project-briefs.json` adheres to this formal structure:

```
{
  "id": "brief-ab-experimentation",
  "title": "Bayesian A/B Experimentation Engine",
  "domain": "ML Engineering",
  "estimatedHours": 29,
  "difficulty": "intermediate",
  "coveredSkills": ["Python", "NumPy", "SciPy", "FastAPI", "SQLite", "Docker", "Statistics"],
  "problemStatement": "Fixed-horizon frequentist experiment evaluation forces product teams to wait...",
  "useCases": "Feature flagging, recommendation algorithm comparison, search ranking...",
  "problemItSolves": "Frequentist peeking inflates false discovery rates; Bayesian sequential methods...",
  "recruiterSignal": "Understands statistical decision theory, sample efficiency, and REST API design...",
  "architecture": "FastAPI ingestion layer -> SQLite state store -> SciPy beta-binomial posterior...",
  "scaleTargets": "Evaluates 10,000 experiment arms in < 50ms; p99 API response < 15ms...",
  "techStack": ["Python", "FastAPI", "SciPy", "NumPy", "SQLite", "Docker"],
  "verificationRubric": [
    "1. Run `pytest tests/test_bayesian_engine.py` to verify posterior convergence",
    "2. Execute `k6 run loadtest.js` to verify 1,000 QPS with zero 5xx responses"
  ],
  "portfolioBulletFormula": "Engineered sequential Bayesian A/B test evaluator in FastAPI, reducing sample exposure time by 38%.
}
```

3.2 The 4 Quality Sentinels of a Production Brief

Quality Sentinel	Quality Standard	Enforcement & Verification Method
1. Zero Invented Facts	No fake salaries, fake hiring guarantees, or invented benchmarks.	Manual audit checklist; missing data stored as `null` / `unverified`.
2. Defensible Scale Targets	Targets must be achievable on a single developer machine.	No "100M QPS"; realistic targets like "1,000 req/s with p99 < 15ms on localhost".
3. Screener Verification Rubric	Concrete automated test commands a screener can execute.	Must contain reproducible commands (e.g. `pytest`, `k6`, `npm test`).
4. Google XYZ Bullet Formula	Resume bullet must follow "Accomplished [X] measured by [Y] by doing [Z]".	Must contain quantifiable metric tags and active engineering verbs.

3.3 Greedy Set-Cover Project Bundler Algorithm (`bundleBriefs.ts`)

When a candidate has open skill gaps $U = {s_1, s_2, dots, s_n}$:

- At each step, calculate marginal coverage: $M(b) = |b. ext{coveredSkills} cap U|$.
- Calculate efficiency: $E(b) = M(b) / b. ext{estimatedHours}$.
- Select brief b^* maximizing $E(b^*)$.
- Update $U leftarrow U setminus b^*. ext{coveredSkills}$ until all gaps are covered.

Chapter 4: The Complete Quality & Scoring Engines

4.1 ATS Ingestion & Layout Quality Engine (`src/engine/ingestion/`)

- **Multi-Format Parsing (`fileParser.ts`)**: Dynamic runtime loading of `pdfjs-dist`, `mammoth`, and `jszip`. Initial load size remains lightweight (1.83 MB).
- **Layout Reader (`layoutReader.ts`)**: Detects multi-column grids, bounding box coordinates, and prevents inter-column line interleaving.
- **Section Segmenter (`sectionSegmenter.ts`)**: Regex classification of 12 standard resume section headings (Experience, Education, Skills, Projects, Certifications, etc.).

4.2 Calibrated 4-Pillar Scoring Mathematics

Pillar Name	Weight	Formula & Core Evaluation Mechanism	Calibration Baseline
1. Hard Skills Coverage	**40%**	`(Evidenced Skills / Target Role Mandatory Skills) * 100`	Mandatory: 3.0, Preferred: 1.5, Mentioned: 1.0
2. Evidence Quality & Depth	**25%**	`(∑ BulletMultipliers) / TotalBullets * 100` L3 (Outcome): 1.0, L2 (Experience): 0.85, L1 (Project): 0.65, L0: 0.35	Regex extracts digits, %, latency ms, QPS, user counts
3. Fresher Readiness	**20%**	Projects (40) + Degree Proof (25) + GitHub Links (20) + CS Theory (15)	Rewards reproducible projects over unverified claims
4. Format Safety	**15%**	Base 100 minus penalties (Multi-column -15, Broken glyphs -10, Tables -15)	Anchor score at 25.0 to prevent demoralizing 0s

4.3 Truth Gate Honesty Engine (`src/engine/tailor/truthGate.ts`)

- **Zero Hallucination Guarantee**: Resume tailoring strictly forbids adding skills not evidenced in the original draft.
- **Parity Checker**: Confirms that bullet point revisions maintain semantic alignment with original accomplishments.

4.4 Pre-Export Formatting Quality Sentinel (`src/lib/tailor/export/formattingChecker.ts`)

- **Broken Glyph Check**: Scans for `&#xFFFD;`, unprintable control characters, and unclosed quotes.
- **Widow & Orphan Lines**: Flags bullet points where the final line contains < 3 words.
- **1-Page Fit**: Checks vertical page height; warns before export if resume overflows standard 1-page bounds.

Chapter 5: Data Models, Storage Adapter & Zod Schemas

5.1 Storage Architecture & Quotas

Store Name	Key Prefix	Quota Cap	Quarantine & Eviction Behavior
Candidate Profile	`hireflow:v4:profile`	5 versions	Prompt to replace or delete when 6th added
Scan History	`hireflow:v4:scan_history`	60 scans	Ring buffer: oldest scan automatically evicted
Saved JDs	`hireflow:v4:saved_jds`	40 JDs	Deduplicated via SHA-256 hashText(); cap prompt
Saved Roles	`hireflow:v4:saved_roles`	50 roles	Cap prompt when limit reached
Local Feedback	`hireflow:v4:feedback`	500 notes	Oldest evicted; invariant: zero scoring mutation
Brief Progress	`hireflow:v4:brief_progress`	30 briefs	Progress status and resume ledger injection

5.2 Tamper-Evident Backup Format & Checksum Verification

Backups are exported as canonical JSON with a tamper-evident checksum (`fnv1a:<hash>`). Upon import, the adapter validates the checksum and Zod schemas, offering **Merge** (combines records) or **Replace** (overwrites existing state) modes.

Chapter 6: Developer's Extension Blueprint ("In Order to Make Changes")

6.1 How to Add or Modify a Subsection in the UI

1. Create or modify the component in `src/features/suite/<tool>/components/` or `src/components/common/` .
2. Ensure it uses standard design tokens (`bg-surface/70` , `border-border/80` , `text-foreground`).
3. Ensure accessibility compliance: visible focus rings (`focus-visible:ring-2`), min 44x44px touch targets.
4. For slide-overs, wrap inside `Drawer.tsx` to automatically gain mobile bottom-sheet ergonomics.

6.2 How to Add or Edit Project Briefs

1. Open `src/data/suite/project-briefs.json` .
2. Add an entry conforming to the 15-field schema.
3. Verify that `portfolioBulletFormula` adheres to the Google XYZ framework.
4. Run the automated dataset validator: `npx vitest run src/data/__tests__/validateData.test.ts` .

6.3 How to Calibrate Scoring Weights & Pillars

1. Edit `ATS_PILLAR_WEIGHTS` in `src/config/scoring.ts` .
2. Note: Changing weights automatically updates `computeConfigFingerprint()` in `src/config/versions.ts` .
3. Update formula descriptions in `docs/metric-inventory.md` and `src/types/explain.ts` .

6.4 Running the Automated Quality Gates

```
# 1. Typecheck the entire codebase
npm run typecheck

# 2. Verify zero network egress / telemetry leaks
npm run privacy:check

# 3. Run all 106 automated test suites (738 passing tests)
npx vitest run

# 4. Build and verify production bundle size budget
npm run build
npx tsx scripts/check-bundle-size.ts
```